PORTFOLIO RESEARCH PAPER

HORMUZ: WAR-RISK COST AND THE ECONOMICS OF TRANSIT

What a supertanker pays to cross the Strait of Hormuz, what the bypass costs instead, and which freight rates and scenario weights decide between them

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Facts and figures as of 4 October 2026

I. BOTTOM LINE

$4.28
expected war-risk premium per barrel, $120M hull, 40/35/25 weights

$9.01
bypass cost per barrel (Yanbu, SUMED, Cape; July fixture)

$4.73
direct freight per barrel above which the bypass wins

13% to 28%
of pre-war flow the bypass pipelines can replace

Transit through Hormuz now turns mainly on war-risk pricing, and barrel supply matters less. On our base case (40% managed partial closure, 35% escalation, 25% durable reopening) the expected war-risk premium for one VLCC transit is about $4.28 a barrel on a $120M hull, roughly 4.4% of cargo value at a probability-weighted Brent of $96.5.

A bypass through Yanbu, the SUMED pipeline and the Cape cost about $9 a barrel in July and has since become dearer. Transiting therefore wins whenever direct freight is below about $4.73 a barrel. That held in normal markets ($1.22 to $2.58) and fails in the market we have now ($10.64 on 23 July). The practical answer is capacity-constrained: send barrels around the strait where the bypass has room, which is only 13% to 28% of the pre-war flow, and price transit for the rest.

Prices have run ahead of the official forecast. EIA's September outlook keeps Brent near $90 for the rest of 2026 and reaches $74 in 2027 only if flows normalise, but it was finished before the 10 to 12 September attack on the Saudi East-West pipeline. EIA spot Brent has since run between $109 and $131 (9 to 29 Sep), with front-month prints near $104 to $108.

Volumes are higher than the headline closure suggests. Estimates that include AIS-dark and ship-to-ship movements put crude leaving the Gulf at 11.8 to 12.8 million b/d, with all liquids near 13.5, against only 5 to 7 million b/d of openly tracked crude. On that evidence the strait looks partly open under military protection. The market cost of that flow is still high: Rory Johnston puts the all-in cost of getting those barrels out at $30 to $40+ a barrel, in line with the roughly $18 to $30 a barrel implied by mid-September Baltic TD3C earnings. The expected-value result at normal freight stands. At today's freight, the verdict runs the other way wherever the bypass has room.

> **What limits this paper.** The scenario weights and the premium mapping are our own judgment. The newest war-risk quotes we found (Reuters via OilPrice, 24 Sep, at 6% to 9% of hull value; Lloyd's List at up to 10%) bracket our 7.5% anchor but are secondary. The newest primary broker quote is Marsh's of 22 July (7.5% to 10%), and we have not found a primary current-level quote since. Section X lists every open gap.

II. SITUATION

The strait closed on 28 February 2026 when the US and Israel struck Iran, reopened briefly under a June memorandum, and has been effectively shut again since early July. Rows below sourced to EIA, IEA or the Saudi Press Agency come from those bodies' own pages. The rest come from press or trackers and are graded secondary.

Date (2026)	Event	Source
28 Feb	US and Israeli strikes on Iran; Iran closes the strait to foreign shipping	Press
Apr	Brent averages about $117 for the month; daily spot high of $138.21 on 7 Apr	EIA, 7 Jul
18 Jun	US-Iran memorandum of understanding; traffic briefly reaches about 20 vessels a day	EIA, 7 Jul; Al Jazeera, 27 Aug
1 Jul	Brent falls below $70, near the pre-war level	Press
Early Jul	Agreement breaks down after attacks on commercial vessels; strait effectively closed again	Press
19 Jul	Houthi attacks resume in the Red Sea; Yanbu loadings fall about 40%, from 5.2 to 3.1 mb/d	Windward, Jul
Aug	Brent averages $91; global stocks down about 400 million barrels year to date	EIA STEO, 9 Sep
9 Sep	EIA sees Middle East output below pre-conflict levels until Q2 2027, Brent near $90 in 2H26 and $74 in 2027 (forecast finished 3 Sep)	EIA STEO, 9 Sep
10 to 12 Sep	Drones hit the Saudi East-West pipeline (about 4 mb/d) and shut it; 5 to 7 days of stored crude at Yanbu	Saudi Press Agency, 12 Sep (primary); Al Jazeera, 11 Sep
11 Sep	IEA: more than 10 mb/d of Gulf output shut in; Gulf crude exports about 13 mb/d in August, nearly half pre-war; stocks down 507 million barrels since February	IEA OMR, 11 Sep
20 Sep	About 1 vessel transits, against 85 to 100 a day before the war	Tracker briefs
22 Sep	Pipeline restarts at a low rate; seven supertankers (14 million barrels) load at Gulf terminals over the weekend	The National (Reuters, satellite data)
23 to 26 Sep	Tanker Trackers puts crude exiting the US blockade line at 13 mb/d, doubled in under a month, of which about 10.35 mb/d transits the strait (its own X post, 23 Sep). Johnston estimates 13.5 mb/d clearing the strait at an all-in $30 to $40+ a barrel	Tanker Trackers (opened); Johnston via Fortune, 26 Sep (relayed)
24 to 25 Sep	Iran offers to reopen within seven days if the US meets its conditions; its security chief says the strait stays shut until all are met	Straits Daily Brief; Fortune
26 Sep	Trump rejects the offer as unacceptable; the WSJ reports he privately expects to resume bombing after the midterms	Fortune, 26 Sep; Yahoo Finance (relayed)
28 Sep	Front-month Brent $105.76, $107.34 intraday in Asia trading; Iran will not ease its conditions	Trading Economics; CNBC (excerpt)
2 Oct	US orders a third carrier group and 2,000+ Marines to the region; G7 agrees a 100 million barrel IEA stock release	Al Jazeera, 2 and 3 Oct

Table 1. Timeline, 28 February to 2 October 2026.

Brent prints in the timeline are mostly front-month or press quotes, and the EIA spot series ran 10% to 20% higher in September. The pipeline outage lasted about 10 to 12 days, well inside the 3 to 6 week repair estimates published right after the attack, and full restoration is put at several weeks. The fall in Brent on the restart did not hold: front-month prints were back at $104 to $108 within days, and EIA spot closed at $113.96 on 29 September,

far above EIA's $90 path. We left out a 24 September quote of $96.84 because it conflicts with two same-day sources. Iran's seven-day offer is a negotiating position, since its own security chief contradicted it the same day.

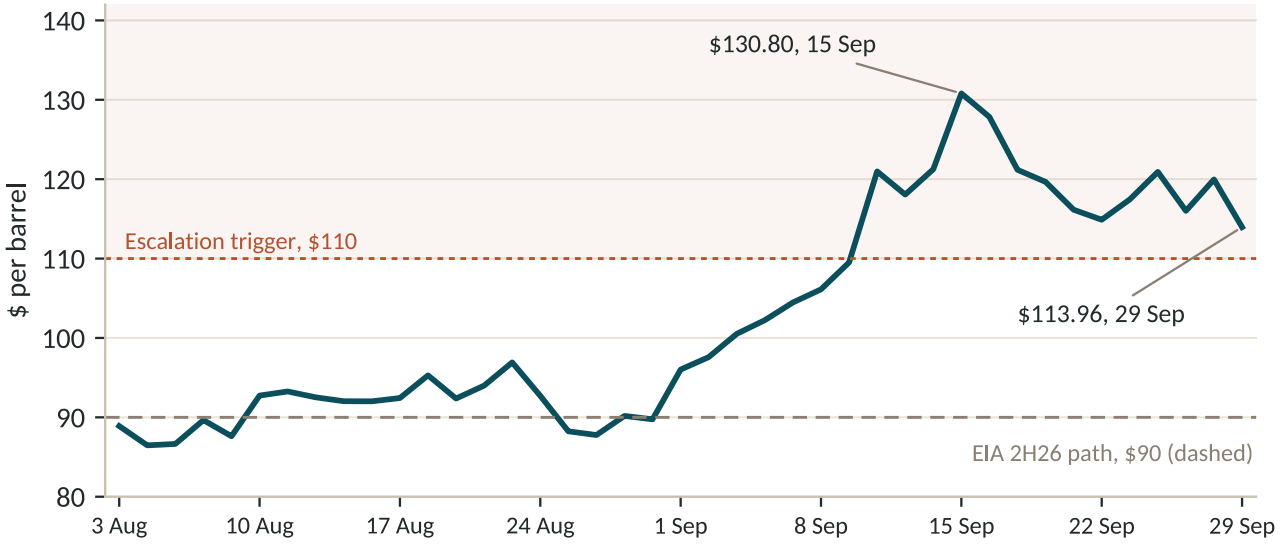


Figure 1. EIA Brent spot, $ per barrel, daily, 3 Aug to 29 Sep 2026. Spot closed above $110 for 14 sessions, 10 to 29 Sep; the September mean through the 29th is $114.08 over 21 sessions, against $91.08 in August. Source: EIA daily series (also on FRED).

III. FLOWS AND BYPASS CAPACITY

Before the war about 20 million barrels a day passed through the strait, and the pipelines around it can replace only 13% to 28% of that. The two agencies disagree on how much pipeline capacity was spare (2.6 versus 3.5 to 5.5 million b/d), and that gap matters more for the transit decision than any price forecast.

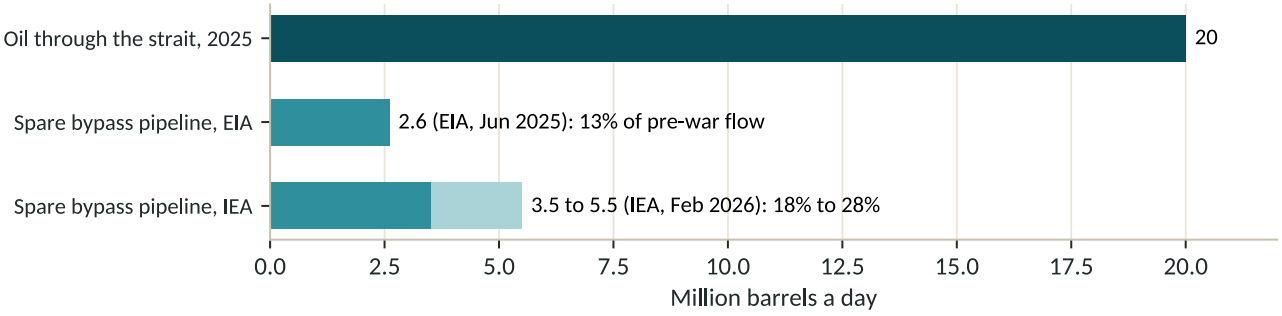


Figure 2. Pre-war flow against spare bypass pipeline capacity. Sources: IEA (Feb 2026), EIA (16 Jun 2025).

Item	Figure	Source
Oil through the strait, 2025	20 million b/d, about 25% of world seaborne oil trade (crude 14.95, products 4.93)	IEA, Feb 2026
LNG through the strait, 2025	About 112 bcm, nearly 20% of world LNG trade	IEA, Feb 2026
Vessel transits per day	About 100 before the war; about 5 from 15 Jul to 23 Aug 2026	Al Jazeera, 27 Aug
Gulf crude exports, four producers	About 17 million b/d in 2025 and about 9 in Aug 2026, down 47%	Al Jazeera, citing Reuters and Kpler
Gulf crude exports, all producers	About 13 million b/d in Aug 2026, nearly half the pre-war level	IEA OMR, 11 Sep
Direct crude flow through the strait	2.2 million b/d average (Kpler)	Al Jazeera, 27 Aug

Table 2. Flow and capacity benchmarks.

The figures reconcile once the IEA's September report is added. Direct flow plus the highest bypass estimate is 7.7 million b/d, which leaves a 12 to 15 million b/d shortfall against the 20 million pre-war figure. The IEA reports more than 10 mb/d of Gulf output shut in and refined product and LPG exports 3.7 mb/d below February, which adds to about 14 mb/d and sits inside that range. The 5 to 7 mb/d disruption quoted by Al Jazeera looks like the outlier, and we cannot say why. The IEA's and Al Jazeera's export levels differ (13 versus 9 mb/d) but both show a fall of about 45%. The likely reason is that the IEA counts all Gulf producers while Al Jazeera lists four, though neither source says so.

How much is actually moving

Newer estimates close most of the gap once the measures are separated (Figure 3). Visible AIS crude, per Kpler, was 49.2 million barrels in the week before 20 September (about 7 million b/d) and 33.7 million in the first six days of the following week (about 5.6 million b/d). Crude including dark and ship-to-ship volumes was 11.8 million b/d on a 7-day average to 25 September (LVision) and 12.8 million b/d on 28 September (Kpler). All liquids including LPG and NGLs was just above 13.5 million b/d (Commodity Context).

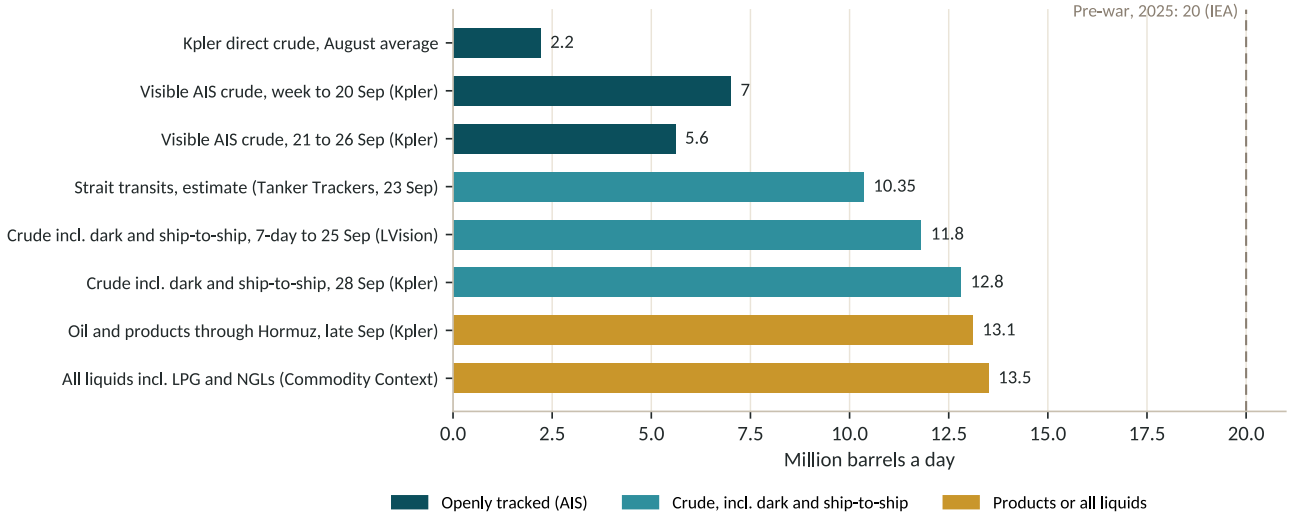


Figure 3. Hormuz flow estimates by what they measure, million barrels a day, late September 2026. LVision and Commodity Context reach us through a Substack summary we opened in full; the originals were not opened. The Kpler 13.1 oil-and-products figure is relayed by roic.ai (30 Sep).

Two figures need careful reading. Tanker Trackers' 13 million b/d, measures crude leaving at the US blockade line, a wider count than strait transits. The post estimates 10.35 million b/d crossing the strait and about 2.65 million b/d leaving from Fujairah and Mina al-Fahal. Rory Johnston's 13.5 million b/d (relayed by Fortune, 26 Sep) sits at the all-liquids tier. Neither is visible AIS crude. The all-in tiers are five to six times Kpler's 2.2 million b/d direct crude average for August, and openly tracked crossings are still tiny: Kpler counted two on 21 September against about 125 large vessels a day before the war.

Flow estimates moved up again on 29 to 30 September. Kpler's figure for the last week of September is 13.1 million b/d of oil and products through Hormuz, about 80% of its 17.1 million b/d pre-war benchmark. JPMorgan puts Middle East crude exports, including bypass routes, at 17.5 million b/d, 98% of pre-war, against only 58% for refined products (roic.ai and Nairametrics, 30 Sep). Both are secondary reports. JPMorgan warns that more crossings do not mean the security environment has improved. These figures make "partly open" a cautious description for crude, though not for fuels. They do not move the scenario weights, which price war-risk premium and escalation risk and leave volume out, and no quote we found has fallen. Kpler's 17.1 million b/d pre-war benchmark is also lower than the 20 million b/d all-liquids figure used elsewhere in this paper, so shares of pre-war flow are not comparable across sources.

Saudi Arabia's own crude shipments averaged 5.28 million b/d in September to the 24th, the fastest pace since February, according to Bloomberg tanker tracking summarised by Briefs.co (the Bloomberg original is paywalled). About 90% of Saudi crude left via the Gulf before the war, and that share fell to zero in April and May. The Briefs.co piece lists the East-West pipeline as offline, which conflicts with Reuters-based reports of a restart on 22 September; we take the restart as confirmed by several outlets. The split between strait and pipeline for the month is unknown.

IV. ACTORS AND INTERESTS

No actor has yet judged the cost of the status quo higher than the cost of a deal, which is why our base case is persistence. The leverage and constraint columns are our reading of the public record and carry no sources.

Actor	Wants	Leverage	Constraint
Iran	Sanctions and blockade relief before reopening; says the strait stays shut without major US concessions	Can raise insurance cost with a few attacks, without a full blockade	Its own ports are under a US blockade; exports and currency are under pressure
United States	Free navigation and lower oil prices	Naval escorts, blockade enforcement	Escorts do not fix insurance; renewed violence pushes Brent up
Gulf exporters (Saudi Arabia, UAE, Iraq, Kuwait)	Volume out	Bypass pipelines and Red Sea routes	Bypass capacity is below pre-war flow; the Red Sea route is also contested
Asian importers (China, India, Japan, Korea)	Continuity of supply	Buy Iran-approved cargoes, draw stocks	Inventories fall through 2026 (EIA)
Insurers and shipowners	Priced risk	Premiums decide who sails	Loss of a hull or crew is the tail they cannot absorb

Table 3. Actors. Iran's and the US positions draw on Al Jazeera, 10 and 12 Aug (opened in full, 4 Oct).

V. SCENARIOS

We weight a managed partial closure at 40%, escalation at 35% and a durable reopening at 25%. The probabilities are our judgment and meant to be challenged. The evidence is base-rate evidence: the April ceasefire and the June memorandum each reopened the strait and then broke down within weeks, and Lloyd's List counts about 70 war-risk claims and roughly $2bn since the war began, with IMO counting 72 attacks on ships since March. Escalation has happened repeatedly. The reopening weight sits close to market pricing: Kalshi put about a 34% chance on normal traffic by 31 December in late August, on a stricter definition than ours, and Polymarket puts a US-Iran nuclear deal by 31 December at 15% (28 to 29 Sep).

Scenario	Probability	Brent ($/bbl)	Rests on
Managed partial closure to Q1 2027	40%	90	EIA 2H26 forecast. Escorted convoys and bypass routes keep some flow; premiums stay high
Escalation into Gulf infrastructure or the Red Sea	35%	120	Our illustration. A strike on export terminals or a wider shipping war; press forecasts range from $85 to above $140
Durable reopening by December	25%	74	EIA 2027 forecast, reached earlier. A signed deal that holds, mine clearance and insurer confidence returning

Table 4. Scenario weights and Brent anchors.

The probability-weighted Brent is $96.5 (0.4 x 90 + 0.35 x 120 + 0.25 x 74), $6.5 above EIA's $90 for 2H26 because the escalation case costs more than the reopening case saves. The anchors sit on different horizons (2H26 for the first two, 2027 for the reopening case), so $96.5 is a rough guide. The 90 and 74 come from EIA's Short-Term Energy Outlook of 9 September 2026; the 120 is ours. All three anchors are stale against September prices (EIA spot reached $138.21 on 7 April and $130.80 on 15 September, both above our escalation figure), and EIA's next outlook (6 Oct) postdates this paper, so we keep its published figures. For scale, an illustrative rebase to 105, 130 and 74 would give $106.0 at these weights ($103.5 at 50/25/25). The premium per barrel does not depend on Brent, only its share of cargo value, which stays near 4.0% either way.

Why the weights moved on 29 September

We held 50/25/25 until 29 September and now publish 40/35/25, moving 10 points from managed closure to escalation. Two things argued for it. Trump rejected Iran's offer on 26 September, and the WSJ reports he privately expects to resume bombing after the midterms on 3 November (via Fortune and Yahoo Finance; the WSJ original is paywalled). And the EIA daily series shows our escalation trigger, a close above $110 for a week, has fired on spot: fourteen straight sessions above $110 from 10 to 29 September. It did not fire on futures. We therefore rest the change mainly on the event base rate, with the trigger as added support. The counter-argument is that a rule this mechanical, on one series, right after it fires, risks overfitting; a reader who weighs it that way can use 50/25/25 directly.

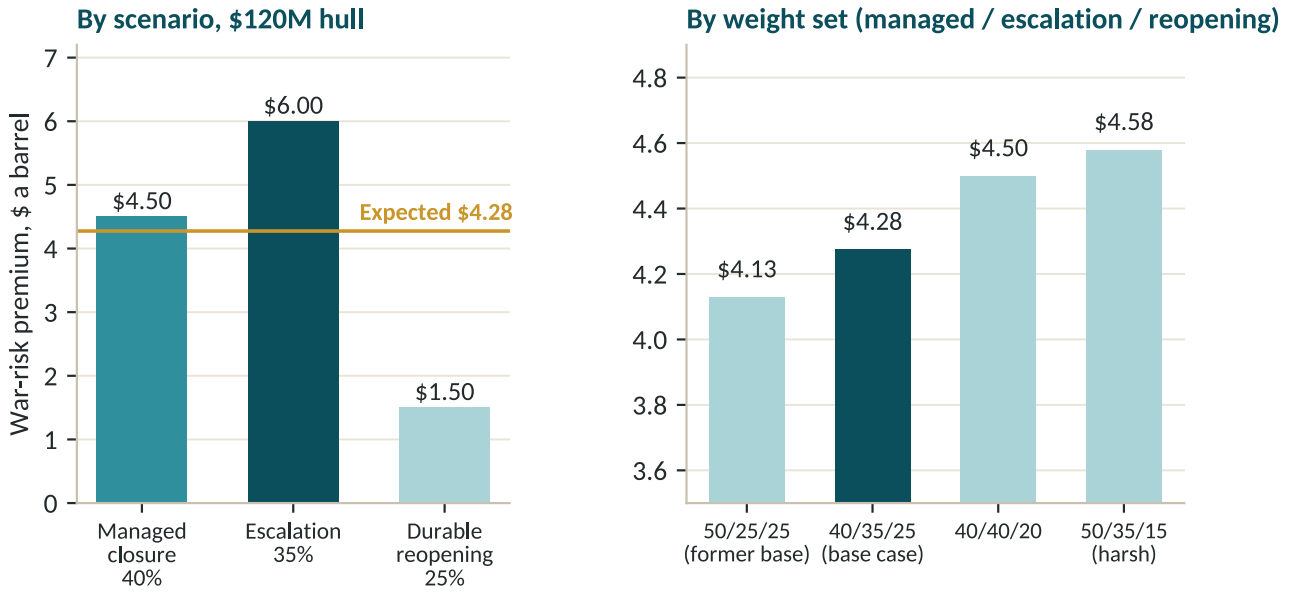


Figure 4. Left: premium cost per barrel by scenario (7.5%, 10% and 2.5% of a $120M hull, 2.0 million barrel cargo). Right: expected premium under four weight sets.

Weights (managed / escalation / reopening)	Expected premium	Weighted Brent	Transit margin at normal freight, $ a barrel
50/25/25 (former base, now stress test)	$4.13	$93.5	$2.31 to $3.67
40/35/25 (base case)	**$4.28**	**$96.5**	**$2.15 to $3.51**
40/40/20 (further five-point shift)	$4.50	$98.8	$1.93 to $3.29
50/35/15 (harsh test)	$4.58	$98.1	$1.85 to $3.21

Table 5. Stress tests. All four keep the verdict at normal freight intact.

Two events since 2 October cut in opposite directions. The US is sending a third carrier strike group and a Marine expeditionary unit with more than 2,000 Marines, with three carriers and two amphibious groups due around Iran by the end of November; officials say the aim is to keep strike options open whatever diplomacy delivers (Al Jazeera, 2 Oct). That supports the escalation weight. Separately, the G7 agreed on 2 October to a coordinated release of 100 million barrels through the IEA over four months, with a substantial diesel release in the first 20 days, and an analyst quoted by Al Jazeera expects only short-lived relief (Al Jazeera, 3 Oct). A stock release moves the price level and leaves the war-risk premium and scenario weights alone. Front-month Brent settled up more than $4 on 1 October and fell back to about $99 to $102 on 2 October (Trading Economics, a live aggregator, so indicative). We hold 40/35/25.

Where the evidence points now. The signals are mixed. Toward reopening: the pipeline restarted on 22 September after 10 to 12 days, faster than repair estimates; seven supertankers loaded at Gulf terminals the weekend before; Saudi shipments reached 5.28 mb/d in September, the fastest since February. Toward persistence or escalation: front-month Brent fell to $97.81 on the restart and was back at $104 to $106 within two days; Iran's offer was contradicted by its own security chief; visible transits were about one vessel a day on 20 September; and the pipeline is at a low rate with full restoration weeks away. If the pipeline keeps ramping, the next revision is more likely to lower the escalation weight than raise it, but the September attack showed how quickly that can reverse.

Triggers for revising the weights. These thresholds are ours and should be tested. Move 10 points from escalation to reopening if Yanbu loadings return above about 4.5 million b/d for two weeks and transits exceed 30 a day for five straight days. Move 10 points from reopening or the base case to escalation if a second strike hits Gulf export infrastructure, if Brent closes above $110 for a week, or if a broker quote comes in above 10% of hull value. Until one of these fires, or the 6 October EIA outlook changes the picture, we hold. A Polymarket deal price above 30% would also move us back toward reopening.

VI. LEADING INDICATORS

Four observable signals should move the scenario weights before the price does. The thresholds are proposals to test against history.

Signal	Reopening if	Escalation if	Latest reading
Daily transits (IMF PortWatch)	Above 30 a day for 5 straight days	Under 5 a day for 10 straight days, or an attack on a convoy	Mean 4.3 a day, 20 Jul to 27 Sep; peak 10. Trigger has not fired
War-risk premium (broker quotes, JWC notices)	Below 2.5% of hull value	Above 10% of hull value	7.5% to 10% (Marsh, 22 Jul); no new current level since
Brent spot against EIA's $90 path	Under $80 for two weeks	Over $110	$113.96 (29 Sep). Above $110 for 14 sessions; trigger has fired on spot, not on futures
Bypass and Gulf loadings (Windward, satellite data)	Yanbu recovers toward the 5.2 mb/d seen before 19 Jul	Yanbu stalls below 3 mb/d, or a second strike hits Yanbu or Fujairah	Pipeline restarted 22 Sep at a low rate; Saudi shipments 5.28 mb/d in September

Table 6. Leading indicators and thresholds.

A recovering Yanbu shows the bypass is repaired, which lowers escalation risk. A durable reopening shows up as Gulf loadings and transits rising while Yanbu's share falls. Kalshi's normalisation contract sets its bar at a 7-day PortWatch average above 60 transit calls, twice our 30-a-day trigger, and priced about a 34% chance of that by 31 December in late August (not refreshed since). We use Kalshi and Polymarket only to cross-check the reopening weight.

Transit counts

We pulled the daily Strait of Hormuz series from the IMF PortWatch feature service for 20 July to 27 September, 70 days, and on 4 October checked every row against the raw service response. AIS-visible transit calls averaged 4.3 a day over the window, with a peak of 10 (8 Aug) and no day at or above 30. The average was 4.7 in August, 3.7 for 1 to 27 September and 3.1 for the seven days to 27 September; tankers were 0 to 3 a day in September. For scale, the series ran 6 to 17 a day on 8 to 19 July and peaked at 50 and 51 on 24 and 25 June during the reopening. Kalshi's bar is a 7-day average above 60, so this series sits at about 5% of it.

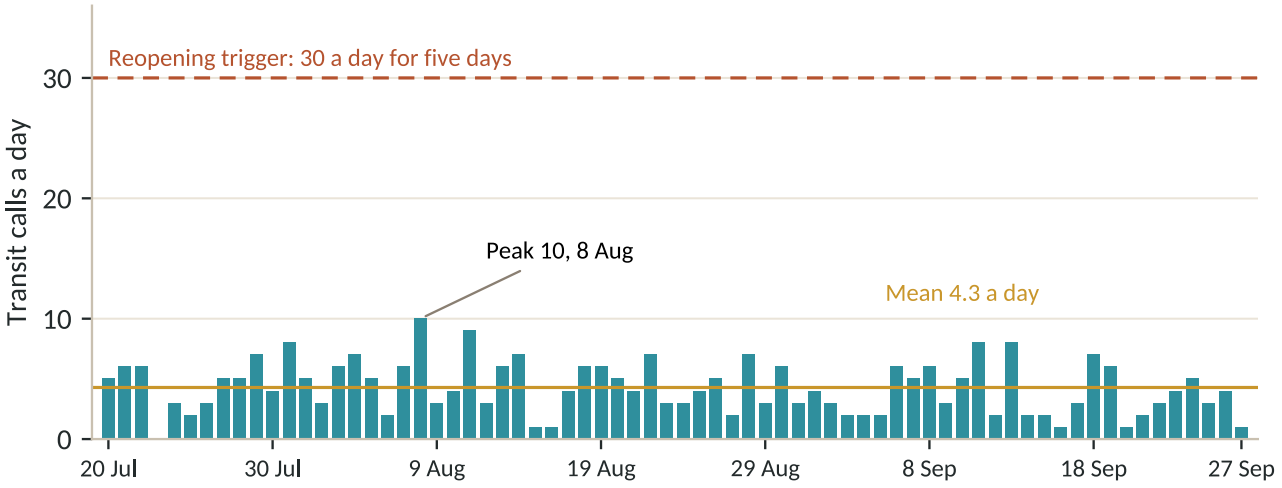


Figure 5. AIS-visible Hormuz transit calls per day, 20 Jul to 27 Sep 2026. Source: IMF PortWatch, Daily_Chokepoints_Data, checked against the raw service response on 4 Oct.

The conflict in the record

PortWatch shows AIS-visible transits near zero through 27 September and tonnage near zero through 20 September, with no sign of the recovery that 10-plus million b/d of escorted flow would need if any material share of it were AIS-visible. The two do not reconcile, and the gap is too large to be rounding. Either dark shipping now accounts for most of the flow, which would itself belong in the recommendation, or the escorted-flow estimates are too high. We cannot say which. Counts from Kpler and Windward on some late-September days run well above PortWatch (Windward 42 on one Sunday, Kpler 40 the next, per a secondary summary), so the sources measure different things.

A mechanism for part of the gap is documented, though not sized. Windward reported on 24 September a ship-to-ship relay pattern: an intermediary vessel receives cargo inside the Gulf, at Basrah and Kuwait anchorages, then transits Hormuz and delivers it outside, at Fujairah and Shinas, so no single AIS track shows an end-to-end crossing. Their case study is one 40,000-dwt product tanker with 15 counterparties and 4 confirmed transfers over 90 days. Separately, Windward reported on 17 September that of 77 vessels Iran has blacklisted for evading its toll regime, 61% still broadcast AIS at some point in September. Both are small-sample case studies, so they explain how barrels could cross undetected without saying how many do. They raise our confidence that some of the claimed flow is real and AIS-dark, without closing the gap to PortWatch.

Spot versus futures

The EIA daily Brent file (data through 29 Sep, also on FRED) is materially higher than the press prints this paper first relied on: Reuters via Daily Caller had $107.55 on 14 September against EIA's $121.25, and The National had $97.81 on 22 September against EIA's $114.89. The highest Dec-contract close we saw for 11 to 29 September was $108.75 (15 Sep). The likely cause is that press quotes are front-month or futures prints in steep backwardation, though sources disagree and we treat that as a hypothesis. Where series conflict, spot governs for cargo value and for the trigger. The 28 September front-month prints ($105.76, $107.37) sit $12.60 to $14.21 below the EIA spot print of $119.97 for that day.

VII. TRANSIT ECONOMICS

The premium

At a 7.5% war-risk premium, one VLCC transit costs about $4.50 a barrel on a $120M hull, roughly 5% of a cargo worth $91 a barrel. We assume a 2.0 million barrel cargo. Hull values of $100M to $140M sit inside the market range of about $71M for a 20-year-old to $151M for a five-year-old, with a newbuilding near $130M (Breakwave, 2 Sep).

For the premium we anchor on Marsh, whose head of marine put Hormuz war-risk rates at 3% to 10% of hull value in mid-July (The National, 17 Jul) and at 7.5% to 10% on 22 July (S&P Global), against 0.25% before the war and 1% to 3% during the June reopening. The newest quote we found is a 24 September Reuters report, relayed by OilPrice, that Hormuz cover is quoted at 6% to 9% of hull value, with about 3% for Saudi-linked tankers calling at Yanbu (up from under 1% in early July). It puts the Hormuz war-risk bill at about $7 million a voyage against at least $100,000 before the war. On 3 September Lloyd's List's David Osler put rates as high as 10% of hull value and 40 to 60 times pre-crisis levels. Our 7.5% managed-closure anchor sits at the middle of 6% to 9%, and 10% is Lloyd's List's stated top.

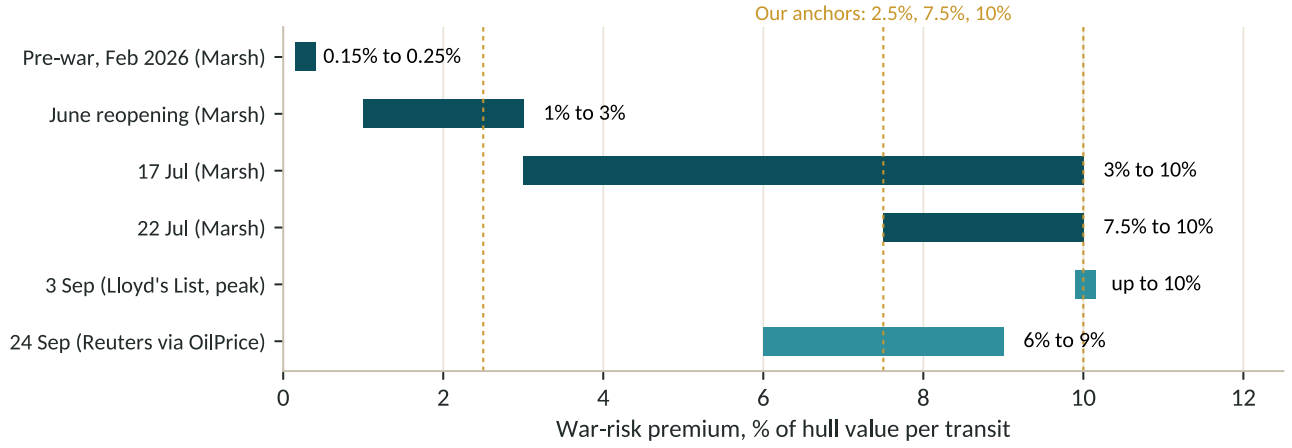


Figure 6. Published war-risk premium quotes against our scenario anchors. Marsh figures were opened on S&P Global (22 Jul), The National (17 Jul), Insurance Business (31 Aug, 2 Sep) and Al Jazeera (23 Jul). Reuters and Lloyd's List figures are relayed by OilPrice and gCaptain.

Premium (% of hull value)	$100M hull	$120M hull	$140M hull
2.5%	$1.25	$1.50	$1.75
5.0%	$2.50	$3.00	$3.50
7.5%	$3.75	$4.50	$5.25
10.0%	$5.00	$6.00	$7.00

Table 7. War-risk cost per barrel (2.0 million barrel cargo).

The same premium weighs less as the price rises, because it is a fixed cost against a larger cargo value.

Premium, $120M hull	Brent $74	Brent $91	Brent $120
2.5%	2.0%	1.6%	1.2%
5.0%	4.1%	3.3%	2.5%
7.5%	6.1%	4.9%	3.8%
10.0%	8.1%	6.6%	5.0%

Table 8. Premium as a share of cargo value.

Breakeven against the bypass

One confirmed fixture in July had a Korean energy major paying $18.5 million to move 280,000 tonnes through Yanbu, the SUMED pipeline and the Cape of Good Hope (Windward, Jul 2026). That is about $9.01 a barrel at 7.33 barrels a tonne (our conversion; $8.70 to $9.20 across the usual range). It is the whole cost of the bypass voyage, so the fair comparison is total against total: direct freight plus the war-risk premium on one side, $9.01 on the other.

$$p^* = (C_{bypass} - F_{direct}) \times Q \div H$$

Here p* is the highest premium, as a share of hull value, at which transiting still costs no more than the bypass. C is the bypass cost per barrel, F is direct freight per barrel, Q is cargo size in barrels and H is hull value. Normal freight is sourced: the five-year average for Gulf to China is $18.91 a tonne, or $2.58 a barrel (Al Jazeera, 23 Jul), and the rate before the June 2025 conflict was about $1.22 a barrel (MEES, 27 Feb).

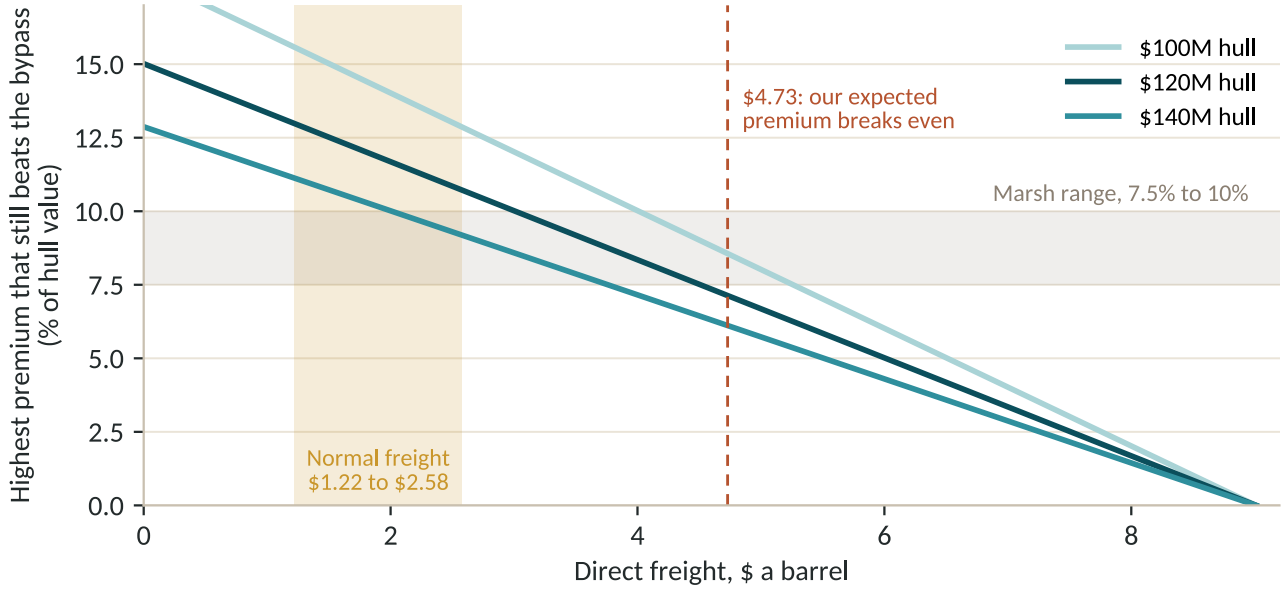


Figure 7. Breakeven premium against direct freight. In normal freight markets transiting beats the bypass at any premium below roughly 9% to 16%, which covers most of Marsh's range.

Direct freight ($ a barrel)	$100M hull	$120M hull	$140M hull
1.22 (before the June 2025 conflict)	15.6%	13.0%	11.1%
2.58 (five-year average)	12.9%	10.7%	9.2%
4.73 (breakeven for our expected premium)	8.6%	7.1%	6.1%
10.64 (spot, 23 Jul)	none	none	none

Table 9. Breakeven premium by freight and hull value.

At the July spot rate the bypass is cheaper even with no premium at all, so freight is the variable to watch.

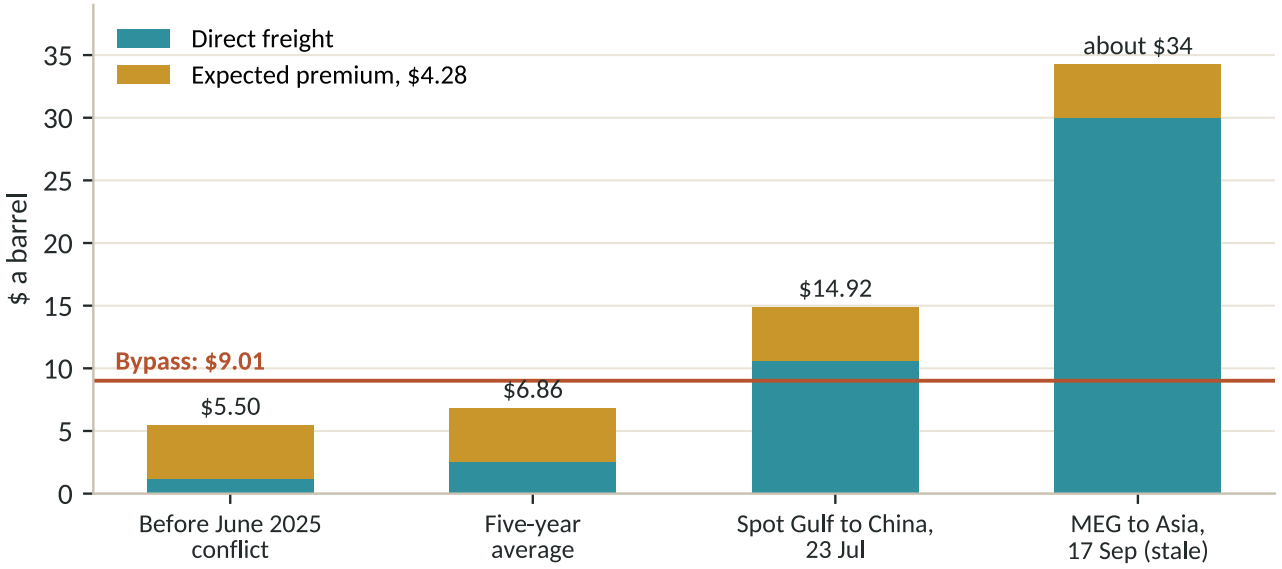


Figure 8. Total transit cost (direct freight plus expected premium) against the bypass. MEG to Asia, 17 Sep, is a stale TankerBrief reading.

Freight basis	Direct freight	Plus premium ($4.28)	Bypass	Transit minus bypass
Before the June 2025 conflict	1.22	5.50	9.01	−3.51
Five-year average	2.58	6.86	9.01	−2.15
Spot Gulf to China, 23 Jul	10.64	14.92	9.01	+5.91
MEG to Asia, 17 Sep (stale)	about 30	about 34	9.01	about +25

Table 10. Transit against bypass, $ a barrel. Negative means transiting is cheaper.

A few things qualify that. The $9.01 is a single July deal and bypass rates have probably risen since, so the July rows are the fairest like-for-like comparison and the September row overstates the gap. Market freight may already include owners' compensation for war risk, so adding the full premium could double count part of it. The bypass cannot carry everything: spare pipeline capacity is 13% to 28% of the pre-war flow, Yanbu loadings fell about 40% after Houthi attacks resumed on 19 July, and the East-West pipeline was down for 10 to 12 days in September and is running at a low rate. Loadings above 5 million b/d before 19 July are consistent with the pipeline having been close to full, which would mean much of the spare capacity had already been used.

September freight

Freight is volatile (Figure 9). Baltic Exchange TD3C (Middle East Gulf to China) was assessed at WS821.11, or $862,150 a day in time-charter-equivalent terms, on 10 September (Xinde Maritime News, quoting Baltic data). Cyprus Shipping News reported WS677.22 and just under $704,000 a day on 9 September. MEES paired about $200,000 a day with nearly $5 a barrel, so scaling linearly puts the September range at roughly $18 to $22 a barrel; the scaling ignores fixed voyage costs and is our own estimate. Rory Johnston's all-in figure of $30 to $40+ a barrel includes insurance and crew cost but excludes the US military, and TankerBrief's $30 from 17 September matches its low end. A 24 September tanker-market note put the shuttle rate across Hormuz itself at about $75 a tonne, roughly $10.2 a barrel (Breakwave); that is a short haul, so it does not slot into the table, but it points the same way. Every reading is far above the $4.73 breakeven. TD3C is thinly assessed while few ships load inside the strait, so it shows direction only.

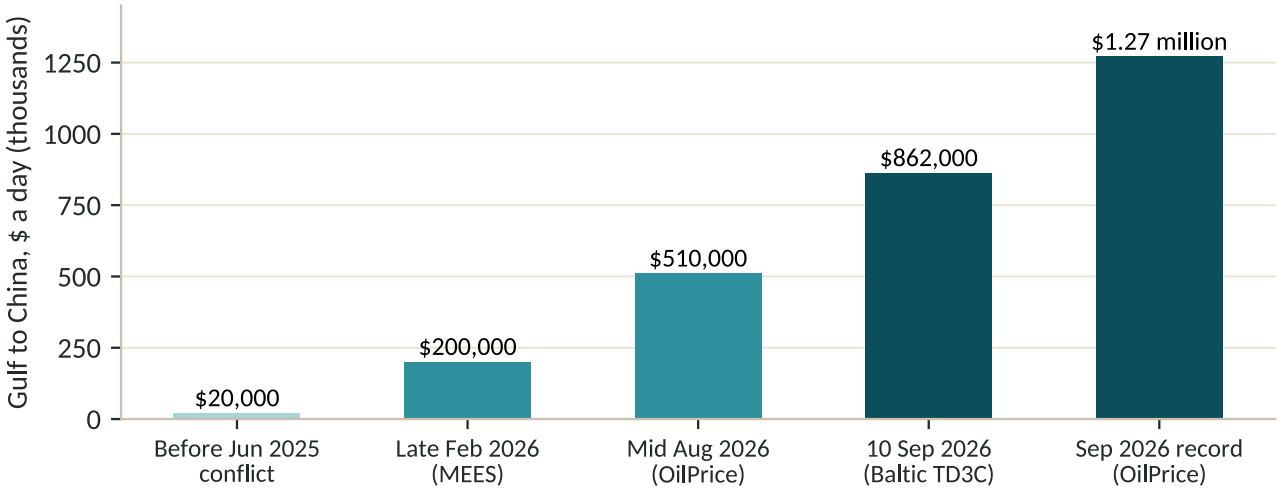


Figure 9. Gulf-to-China VLCC day rates across the readings we found. Sources: MEES (27 Feb), OilPrice (Aug, Sep), Xinde Maritime News (Baltic data, 10 Sep).

The bypass has also become dearer. Yanbu war-risk cover rose to about 3% of vessel value by 24 September from under 1% in early July, which on a $120M hull and a 2.0 million barrel cargo is about $1.80 a barrel against under $0.60 in July, and charters were at least $500,000 a day. The $9.01 July fixture therefore understates the September bypass by more than $1.20 a barrel before any freight change. That narrows but does not close the gap to the $18 to $40 September transit freight.

Breakeven against waiting

A shipper can also wait or transfer cargo at sea instead of paying the premium, so the delay a premium is worth depends on the day rate, and the two figures we found are twelve times apart.

Premium (% of $120M hull)	At about $100,000 a day (Lloyd's List, to 3 May)	At $1.27 million a day (OilPrice, Sep)
3%	36 days	2.8 days
5%	60 days	4.7 days
7.5%	90 days	7.1 days
10%	120 days	9.4 days

Table 11. Delay equivalent to the premium.

At $1.27 million a day, a record spot rate tied to ship-to-ship transfers in the Gulf of Oman, a 7.5% premium equals about 7 days of delay, so transiting beats any longer wait. At $100,000 a day the same premium equals 90 days, and waiting or rerouting is cheaper. The two rates are for different routes and dates, so we do not average them. Waiting is not free either: in September the same Saudi export volume needed 40 VLCCs against 24 in August because of ship-to-ship transfers (OilPrice). The case for transiting strengthens as day rates rise and the bypass fills.

Expected cost across scenarios

Combining the scenario weights with premium costs means mapping each scenario to a premium level. That mapping is our judgment: managed partial closure at 7.5%, escalation at 10% and durable reopening at 2.5%. The 2.5% sits inside the 1% to 3% Marsh reported during the June reopening, and 7.5% and 10% are the bottom and top of Marsh's 22 July range. The probability-weighted premium is 0.4 x 4.50 + 0.35 x 6.00 + 0.25 x 1.50 = $4.275, rounded to $4.28 a barrel, against a weighted Brent of $96.5, about 4.4% of cargo value.

On expected value and normal freight, transiting beats the bypass: $4.28 of premium plus $1.22 to $2.58 of freight is $5.50 to $6.86, against $9.01, a margin of $2.15 to $3.51 a barrel. The margin holds partly because the reopening case pulls the average down. In the escalation case alone, premium plus normal freight is $7.22 to $8.58 on a $120M hull, a margin of only $0.43 to $1.79, and it reverses on a $140M hull at the five-year-average freight ($7.00 plus $2.58 is $9.58). At the July spot freight of $10.64 the bypass wins on expected value by $5.91. Freight sets the verdict before the scenario weights do.

The mapping has two weak points. The newest quote we could read in full is Marsh's 22 July. Marsh executives quoted by Insurance Business on 31 August and 2 September restate 10% as the peak, while a 26 August piece in the same outlet puts cover at 3% to 8%, and the September figures are secondary, so we have no primary current-level quote since 22 July. And 10% is the top of the reported range and could be exceeded, so escalation could cost more than we have modelled. For scale, war-risk premiums peaked at 3.5% of hull value in the 2003 Iraq invasion and fell to 0.25% by early 2004 (Strauss Center, undated page), so a fast fall after a real reopening is plausible.

Quotes and payments can differ. A 28 July trade report says Hormuz cover was quoted at 7.5% to 10% of hull value, sometimes higher, though few shipowners appeared to be buying at that level (i-law, paywalled; read as an excerpt). Lloyd's List reported in March that additional premiums are negotiable and often waived for favoured customers, with safer vessels paying about 1% of hull value or less (excerpt). If premiums actually paid sit below the quotes, our expected premium is conservative; by how much is unknown.

> **A live counter-argument.** The Lloyd's Market Association stated on 23 March that war cover is available and priced, not withheld, citing 88% of the Lloyd's marine war market maintaining appetite for hull war risk and over 90% for cargo. It argued that reduced transits reflect crew-safety judgment by owners and masters (at least 11 crew fatalities, depleted supplies, limited refuge ports), not an insurance constraint. If that framing holds, the premium we model is a real cost and may not be the binding one, and volume would stay low even if the premium fell. We have no evidence the framing has changed since March, and no data to size how much of the current gap it explains. We leave it open.

Sensitivity: what moves the answer

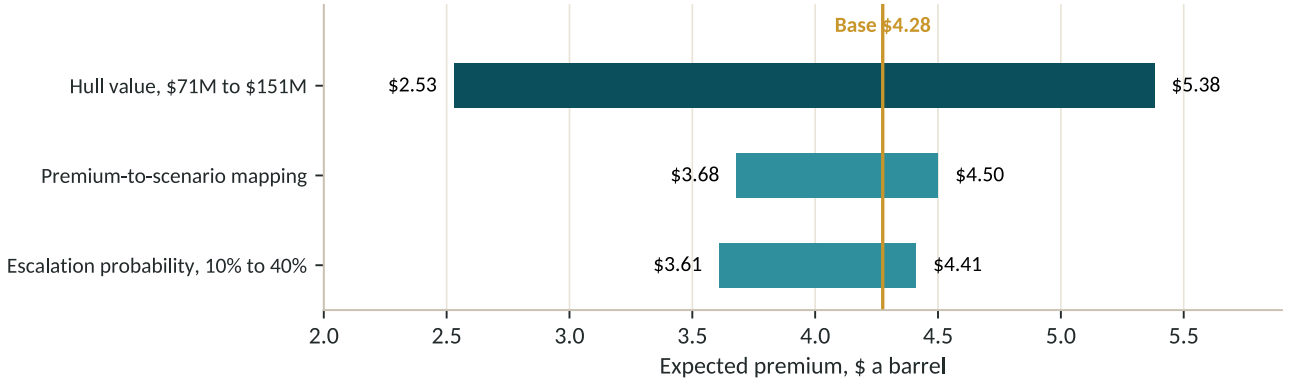


Figure 10. Swing in the $4.28 expected premium from each assumption, holding the others at base case. Mapping tests: flat 7.5%; 2.5, 5 and 10; our 2.5, 7.5 and 10. Escalation probability: other weights rescaled in proportion.

Hull value moves the premium most ($2.85 swing). Signal Ocean put a newbuilding at about $129M and a five-year-old at $138M in May (Seatrade, 8 May), and Breakwave gave $130M, $151M and $71.1M for a newbuilding, a five-year-old and a 20-year-old on 2 September. We keep $120M as the base for a mid-age Gulf VLCC. Age matters because the premium is charged on hull value, so an older ship pays less in dollars at the same rate. The mapping ($0.82) is our own judgment and deserves more scrutiny than hull value. Probability weights ($0.80) now matter about as much as the mapping: an argument that 40/35/25 is wrong does less damage than an argument about the hull.

All three are smaller than the freight market. Direct freight runs from $1.22 to $10.64 a barrel across the readings we found, a $9.42 swing in total transit cost, and it flips the verdict where the others do not. With our expected premium, the verdict flips when direct freight exceeds about $4.73 a barrel.

VIII. EXPOSURE AND MITIGATION

Exposure sits with anyone whose margin depends on Gulf crude or distillates arriving on schedule, and EIA's own numbers show the pressure in products first: its September outlook puts US distillate stocks below 100 million barrels and retail diesel at $5.07 a gallon for 2026.

Exposure	Why it matters	Mitigation to test
Asian refiners and importers	Take most Hormuz cargoes; inventories are falling	Longer-dated supply contracts, non-Gulf grades
Distillate-dependent operators (airlines, logistics)	EIA sees distillate tightness through much of 2027	Fuel hedges sized to the $90 path, reviewed after the 6 Oct outlook
Shipowners with Gulf-trapped vessels	Premiums of 7.5% to 10% of hull value per transit by late July (Marsh)	Charter clauses on war-risk cost allocation
Hospitality and travel businesses	Fuel and airfare pass-through	Scenario budgets at $74, $90 and $120 Brent

Table 12. Exposure map.

IX. RECOMMENDATION

The call depends on which freight market a shipper faces, and capacity decides how much of the flow that choice covers. At normal freight ($1.22 to $2.58 a barrel), transiting beats the bypass on expected value by $2.15 to $3.51 a barrel, with an expected premium of $4.28 against $9.01 for the bypass. At the July spot freight ($10.64) the bypass wins by $5.91. The bypass can carry only 13% to 28% of the pre-war flow, so for a shipper with the flexibility to choose, the order is bypass first where it has room, transit for the remainder, and a premium hedged at the level a broker will actually quote which replaces our 7.5% anchor once a fresher number exists. Spot Brent near $114 (EIA, 29 Sep) and front-month near $106 (28 Sep), against EIA's $90 path, say the market is not yet pricing the reopening case as confidently as our expected-value arithmetic implies.

At September freight the call reverses for any barrel the bypass can carry: our scaling of Baltic TD3C earnings gives roughly $18 to $22 a barrel and Johnston's all-in cost is $30 to $40+, against $9.01 for the bypass, which has itself become dearer since July. Because estimates that include dark and ship-to-ship volumes put crude leaving the Gulf at 11.8 to 12.8 million b/d while visible AIS crude is only about 5 to 7, the practical constraints include escort access, ship-to-ship capacity and freight, along with bypass capacity.

For exposure, Asian refiners and distillate-dependent operators carry the most direct risk and have the least control over the strait, so contract length and non-Gulf sourcing are the mitigations to test first. Shipowners face the transit-versus-bypass decision modelled here. Travel and hospitality exposure is secondary and mostly a pass-through problem, and a scenario budget covers it better than active hedging.

> **What would change this call.** A premium quote well above 10% of hull value, or a higher-value hull, would push the verdict toward the bypass where it has room. A freight rate that settles below about $4.73 a barrel would widen the transit case, and a sustained fall in Yanbu loadings would take the bypass away for the barrels it can no longer carry. A durable ceasefire or a second attack on Gulf infrastructure would move the scenario weights, which matter less than the freight market and the hull value, and about as much as the premium mapping.

X. CONFIDENCE AND OPEN GAPS

Confidence is moderate on volumes, the EIA price path and the freight baseline, and low on the current insurance cost, the September freight level and the September Brent path. Trade shares, pipeline capacity, hull values and normal freight are confirmed on pages we opened. The gaps below are still open.

Gap	Status	How to close
War-risk premium	No primary current-level broker quote since Marsh's 22 Jul (7.5% to 10%). Marsh comments on 31 Aug and 2 Sep restate 10% as the peak. September levels (Reuters 6% to 9%, via OilPrice; Lloyd's List up to 10%) are secondary. The verdict does not need a point quote: transiting beats the bypass at normal freight for any premium up to 13.0% (at $1.22) or 10.7% (at $2.58) on a $120M hull. On a $140M hull the five-year-average threshold falls to 9.2%, so the top of the published range would flip the result there	A current broker quote or Joint War Committee notice
Flow tiers and transit counts	PortWatch rows checked against the raw feed. Kpler's, LVision's and Commodity Context's own notes were not opened; LVision and Commodity Context are secondhand. Visible counts and dark-inclusive estimates conflict and we do not pick a side	The Kpler or LVision notes
Brent in September	Mostly resolved: EIA daily spot is in hand through 29 Sep. Front-month prints are secondary, and spot, front-month and Dec futures differ by $10 to $20	Pull the EIA or FRED spot file the day after the 7 Oct release
Spare bypass capacity	EIA (Jun 2025) says 2.6 million b/d; IEA (Feb 2026) says 3.5 to 5.5	Ask which basis each uses; check operator statements
Bypass cost	One fixture, about $9.01 a barrel (SUMED plus Cape, July)	A second fixture
September freight	TankerBrief's $30 a barrel (17 Sep) is stale; Baltic TD3C is thinly assessed	A dated September Gulf-to-China print
Saudi September exports	The 5.28 mb/d is a Bloomberg figure summarised by Briefs.co; split between strait and pipeline unknown	The Bloomberg original, or Saudi data by route
Relayed reports	WSJ report on post-midterm bombing is relayed by Fortune and Yahoo Finance; WSJ original paywalled. Johnston's 13.5 million b/d and $30 to $40+ is relayed by Fortune. CNBC and CNN carried the first pipeline-attack reports but would not open for us	The originals
Our assumptions	Scenario probabilities, premium mapping, $120 escalation price, 2.0 million barrel cargo, $120M base hull, 7.33 barrels a tonne	Argue them in review; test sensitivity

Table 13. Open gaps and how to close them.

Attack date. The disruption was reported in stages. Al Jazeera, which we opened, dates the drone strike 11 September; Saudi Arabia's Foreign Ministry condemned it in a statement dated 12 September (Saudi Press Agency, primary); The National dates the shutdown to 13 September; and Reuters via Daily Caller reports the sharpest Brent move on 14 September. We use 10 to 12 September as the start, which makes the outage about 10 to 12 days to the 22 September restart.

Resolved since 25 Sep: the volume shortfall (IEA September report), the pipeline repair time (restart on 22 Sep after 10 to 12 days), the hull value range (Breakwave and Seatrade) and the normal freight baseline (MEES and Al Jazeera). We did not use the claim of a 39% loss of global trade, which rests on one encyclopedia-style page, or the $96.84 Brent quote of 24 September. The Congressional Research Service report R45281 blocks automated access, so we relied on the IEA for trade shares.

APPENDIX. SOURCE REGISTER

Status shows what we did with each source. "Opened" means we read the page itself; "relayed" means we saw it only through another outlet; "excerpt" means a search excerpt only.

Source	Used for	Status
Official and primary		
EIA, Short-Term Energy Outlook, 9 Sep 2026	Brent path, inventories, output outlook, diesel and distillate figures	Opened
EIA press release, 7 Jul 2026	18 Jun memorandum, June Brent average, April peak	Opened
EIA, Today in Energy, 16 Jun 2025	Spare bypass capacity (2.6 million b/d)	Opened
IEA, Strait of Hormuz key figures, Feb 2026	2025 flows, trade shares, LNG, bypass capacity (3.5 to 5.5 million b/d)	Opened
IEA, Oil Market Report, 11 Sep 2026	Shut-in output, Gulf exports, product exports, stock draws	Opened
Saudi Press Agency, 12 Sep 2026	Official condemnation of the pipeline attack	Opened (earlier pass)
FRED, Brent spot (DCOILBRENTEU)	EIA daily spot series through 29 Sep	Opened
IMF PortWatch, Daily_Chokepoints_Data feature service	Daily Hormuz transit calls, 20 Jul to 27 Sep	All 70 rows checked against raw response, 4 Oct
Brokers, analysts and data providers		
S&P Global, 22 Jul 2026	Marsh (Marcus Baker): 7.5% to 10% of hull value, up from 1% to 3%	Opened 4 Oct (fetch summary)
Insurance Business, 26 Aug to 2 Sep 2026	Marsh: peak up to 10%, post-February 3% to 10%; 26 Aug: 3% to 8%	Opened 4 Oct (fetch summary)
Windward, Jul 2026	SUMED plus Cape fixture; Yanbu loadings around 19 Jul	Opened; single fixture
MEES, 27 Feb 2026	Gulf-to-China freight, about $200,000 a day (nearly $5 a barrel) against about $20,000 ($1.22) before June 2025	Opened
Breakwave, 2 Sep and Seatrade Maritime, 8 May 2026	VLCC newbuilding and secondhand prices (May figures cite Signal Ocean)	Opened
Breakwave, 24 Sep 2026	Hormuz shuttle freight rate	Opened (earlier pass)
Lloyd's List, VLCC volumes	VLCC day rates to 3 May 2026 (about $100,000)	Excerpt
i-law, 28 Jul and Lloyd's List, 11 Mar 2026	Quotes versus premiums actually paid	Excerpts (paywalled)
Strauss Center, insurance market	Historical premium rules of thumb	Opened; undated page
TankerBrief, 23 Sep 2026	MEG freight about $30 a barrel (17 Sep)	Opened; stale tracker
JWC circular JWLA-034, 29 Jul 2026	Listed areas for hull war risk	Opened; gives no rates
Xinde Maritime News and Maritime News, Sep 2026	Baltic TD3C: WS821.11, $862,150 a day (10 Sep); about $704,000 (mid-Sep)	Opened; Cyprus Shipping News matches
Nick Wade Substack, late Sep 2026	LVision 11.8, Commodity Context 13.5 all liquids, Kpler crossing count	Opened in full; reports the originals
Tanker Trackers, X post, 23 Sep 2026	13 mb/d blockade-line egress; 10.35 mb/d strait transits	Opened 4 Oct
Baird Maritime, 25 Sep	Kpler visible crude volumes, weeks of 13 and 20 Sep	Opened
roic.ai and Nairametrics, 30 Sep 2026	Kpler 13.1 mb/d oil and products; JPMorgan 17.5 mb/d crude, 98% of pre-war	Opened through a summariser; Kpler and JPMorgan originals not opened
Press and aggregators		
Al Jazeera, 27 Aug 2026	Transits, Gulf exports, Kpler direct flow	Opened

Source	Used for	Status
Al Jazeera, 23 Jul 2026	Marsh premium; Gulf-to-China freight ($77.96 a tonne now, $18.91 five-year average)	Opened
Al Jazeera, 10 Aug and 12 Aug 2026	Iranian position, blockade; October Brent futures $84.11 and $89.53	Opened in full 4 Oct
Al Jazeera, 11 Sep 2026	East-West pipeline hit by drones launched from Iraq; stopped as a precaution	Opened 4 Oct (fetch summary)
Al Jazeera, 2 Oct and 3 Oct 2026	Third carrier group, Marines, G7 release (the 3 Oct lede says 100 billion, a typo for 100 million)	Opened in full 4 Oct
The National, 17 Jul and 22 Sep 2026	Marsh range 3% to 10%; pipeline restart, loadings, Brent $97.81	Opened
OilPrice, Aug and Sep 2026	$510,000 and $1.27 million a day; ship-to-ship volumes	Opened
OilPrice, 24 Sep 2026	Hormuz war-risk 6% to 9%, Yanbu about 3%, charters at least $500,000 a day	Opened; relays Reuters, original not opened
OilPrice, 29 Sep 2026	Kpler 12.8 million b/d out of Hormuz; front-month Brent $107.37	Opened
gCaptain, 3 Sep 2026	Lloyd's List (David Osler): about $2bn claims, rates up to 10%; IMO count of 72 attacks	Opened; relays Lloyd's List
Fortune, 26 Sep 2026	Trump's rejection; WSJ report; Tanker Trackers and Johnston figures	Opened in full; posts and WSJ report relayed
Yahoo Finance, 28 to 29 Sep 2026	Polymarket odds; WSJ report relayed ("US officials told the Wall Street Journal")	Opened; WSJ original paywalled
Daily Caller, 14 Sep and Briefs.co, 24 Sep 2026	Pipeline attack and Brent $107.55 (Reuters); Saudi September exports (Bloomberg)	Opened; both relay paywalled originals
ChemAnalyst, 24 Sep, Straits Daily Brief, 24 and 25 Sep, Fortune, 24 Sep	Pipeline restart; Brent $104.67 and $104.37; one transit on 20 Sep; Iran's offer contradicted	Opened 2 Oct
Trading Economics and CNBC, 28 Sep 2026	Brent $107.34 intraday, $105.76; Iran's stance	Excerpts; live-price page not relied on
Investing.com, Brent futures history	Dec-contract closes, 11 to 29 Sep	Opened; aggregator
Octagon	Kalshi normalisation odds (cross-check only)	Opened
Hellenic Shipping News, 29 Mar 2026	Baltic Gulf of Oman alternative route	Excerpt
Foreign Policy, 1 Oct 2026, "Oil Is Again Flowing Out of Hormuz"	Same flow theme as the Kpler and JPMorgan figures	Headline and standfirst only
Not reachable		
Congressional Research Service report R45281; IMF PortWatch and EIA chokepoint web pages (no usable data); EIA daily Brent history page (stops in 1999 in the extract); CNBC pipeline article (access denied); CNN, 11 Sep (refused for legal reasons); AGBI Gulf-China freight piece (access refused).		

Facts and figures as of 4 October 2026. Scenario weights, premium mapping and the $120 escalation price are the authors' assumptions.